

Tonmya_Prescriptions

2026-08-22

Some simple models based on Tonmya sales

What are we doing here? - Just trying to get a grasp on the rate of increase and projecting out if that rate continues. It is not necessarily realistic to take it out as far as projections provide. Nothing here should be regarded as investment advice. I am not a financial analyst or advisor.

scripts Projection - at weekly rate - *linear model*

Call:

```
lm(formula = scripts ~ wk, data = df_scripts)
```

Residuals:

Min	1Q	Median	3Q	Max
-80.943	-32.532	-3.872	29.603	97.007

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-38.1417	14.4025	-2.648	0.0118 *
wk	27.0071	0.6276	43.034	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 44.11 on 37 degrees of freedom

Multiple R-squared: 0.9804, Adjusted R-squared: 0.9799

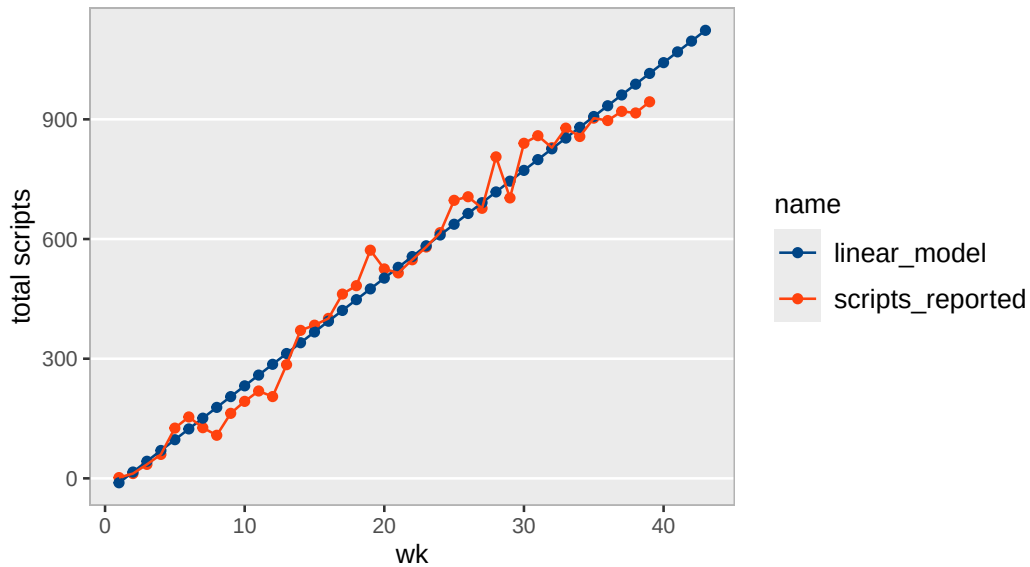
F-statistic: 1852 on 1 and 37 DF, p-value: < 2.2e-16

Scripts, Linear Model

wk	weekly_seq	total reported	linear predicted	annual
1	2025-11-14	2	-11.1	1,404
2	2025-11-21	12	15.9	2,809
3	2025-11-28	35	42.9	4,213
4	2025-12-05	60	69.9	5,617
5	2025-12-12	126	96.9	7,022
6	2025-12-19	154	123.9	8,426
7	2025-12-26	127	150.9	9,831
8	2026-01-02	108	177.9	11,235
9	2026-01-09	163	204.9	12,639
10	2026-01-16	193	231.9	14,044
11	2026-01-23	219	258.9	15,448
12	2026-01-30	205	285.9	16,852
13	2026-02-06	285	313.0	18,257
14	2026-02-13	371	340.0	19,661
15	2026-02-20	384	367.0	21,066
16	2026-02-27	401	394.0	22,470
17	2026-03-06	462	421.0	23,874
18	2026-03-13	483	448.0	25,279
19	2026-03-20	572	475.0	26,683
20	2026-03-27	525	502.0	28,087
21	2026-04-03	515	529.0	29,492
22	2026-04-10	548	556.0	30,896
23	2026-04-17	580	583.0	32,300
24	2026-04-24	616	610.0	33,705
25	2026-05-01	697	637.0	35,109
26	2026-05-08	706	664.0	36,514
27	2026-05-15	677	691.0	37,918
28	2026-05-22	806	718.1	39,322
29	2026-05-29	703	745.1	40,727
30	2026-06-05	840	772.1	42,131
31	2026-06-12	859	799.1	43,535
32	2026-06-19	829	826.1	44,940
33	2026-06-26	878	853.1	46,344
34	2026-07-03	857	880.1	47,749
35	2026-07-10	903	907.1	49,153
36	2026-07-17	897	934.1	50,557
37	2026-07-24	920	961.1	51,962
38	2026-07-31	916	988.1	53,366
39	2026-08-07	944	1,015.1	54,770
40	2026-08-14	NA	1,042.1	56,175
41	2026-08-21	NA	1,069.1	57,579
42	2026-08-28	NA	1,096.2	58,983
43	2026-09-04	NA	1,123.2	60,388

Tonmya scripts with linear model

raw Symphony data



```
[1] 0.7349111
```

Scripts Projection - at weekly growth rate - *exponential model*

Formula: scripts ~ a * (1 + r)^wk

Parameters:

	Estimate	Std. Error	t value	Pr(> t)
a	1.705e+02	1.797e+01	9.488	1.89e-11 ***
r	4.918e-02	3.547e-03	13.866	3.14e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 102.9 on 37 degrees of freedom

Number of iterations to convergence: 11

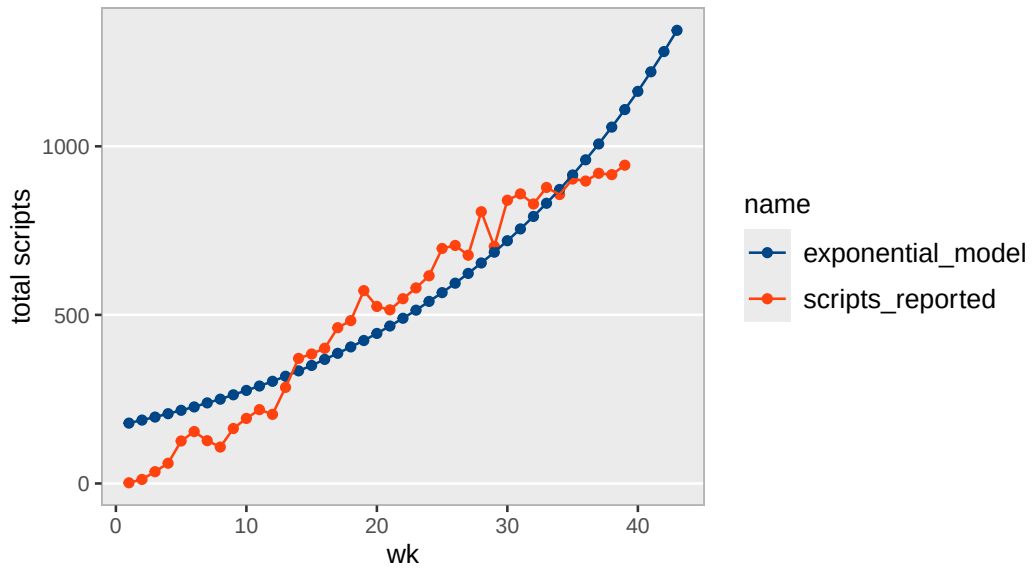
Achieved convergence tolerance: 7.688e-06

Scripts, Exponential Model

wk	weekly_seq	total reported	exp predicted	annual
1	2025-11-14	2	179	9,301
2	2025-11-21	12	188	9,759
3	2025-11-28	35	197	10,239
4	2025-12-05	60	207	10,742
5	2025-12-12	126	217	11,271
6	2025-12-19	154	227	11,825
7	2025-12-26	127	239	12,407
8	2026-01-02	108	250	13,017
9	2026-01-09	163	263	13,657
10	2026-01-16	193	276	14,329
11	2026-01-23	219	289	15,033
12	2026-01-30	205	303	15,773
13	2026-02-06	285	318	16,548
14	2026-02-13	371	334	17,362
15	2026-02-20	384	350	18,216
16	2026-02-27	401	368	19,112
17	2026-03-06	462	386	20,052
18	2026-03-13	483	405	21,038
19	2026-03-20	572	424	22,073
20	2026-03-27	525	445	23,158
21	2026-04-03	515	467	24,297
22	2026-04-10	548	490	25,492
23	2026-04-17	580	514	26,746
24	2026-04-24	616	540	28,061
25	2026-05-01	697	566	29,441
26	2026-05-08	706	594	30,889
27	2026-05-15	677	623	32,408
28	2026-05-22	806	654	34,002
29	2026-05-29	703	686	35,675
30	2026-06-05	840	720	37,429
31	2026-06-12	859	755	39,270
32	2026-06-19	829	792	41,201
33	2026-06-26	878	831	43,228
34	2026-07-03	857	872	45,354
35	2026-07-10	903	915	47,584
36	2026-07-17	897	960	49,924
37	2026-07-24	920	1007	52,380
38	2026-07-31	916	1057	54,956
39	2026-08-07	944	1109	57,659
40	2026-08-14	NA	1163	60,494
41	2026-08-21	NA	1221	63,469
42	2026-08-28	NA	1281	66,591
43	2026-09-04	NA	1344	69,866

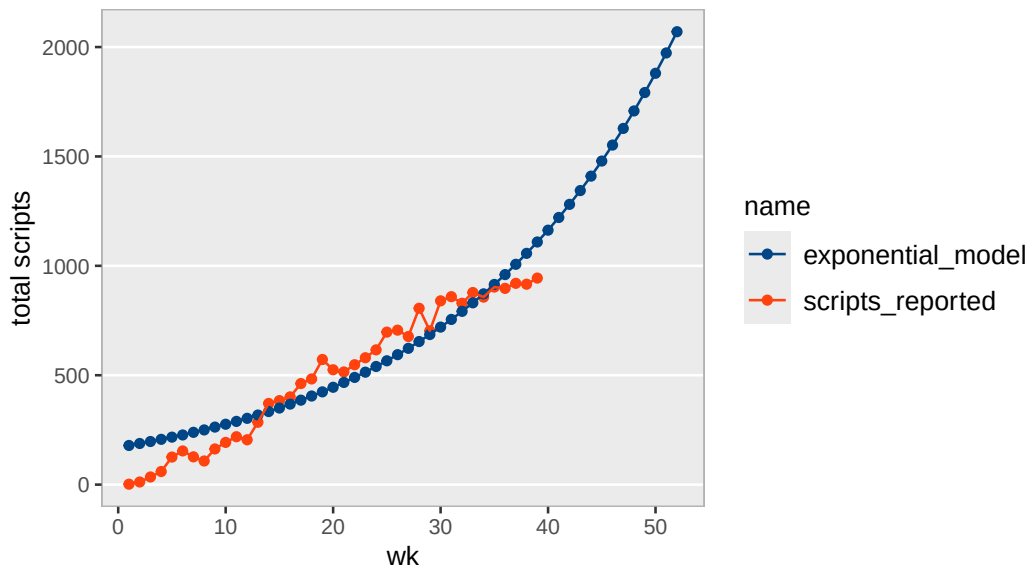
Tonmya scripts with exponential model

raw Symphony data



Tonmya scripts with exponential model

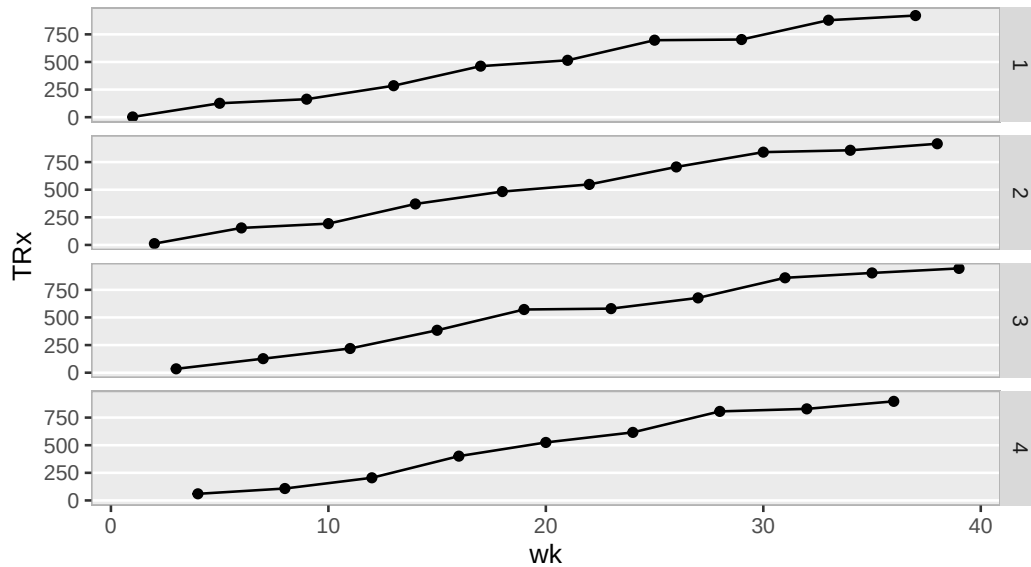
raw Symphony data – for giggles



TRx Total Prescriptions by Weekly Cohorts

Tonmya total prescriptions

broken into week cohorts 1 through 4, raw Symphony data



total / new scripts at weekly rate - *linear model*

Call:

```
lm(formula = total_to_new ~ wk, data = trx_nrx_df)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.063558	-0.033798	-0.000234	0.032789	0.069340

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9528782	0.0125696	75.81	<2e-16 ***
wk	0.0157853	0.0005477	28.82	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.0385 on 37 degrees of freedom

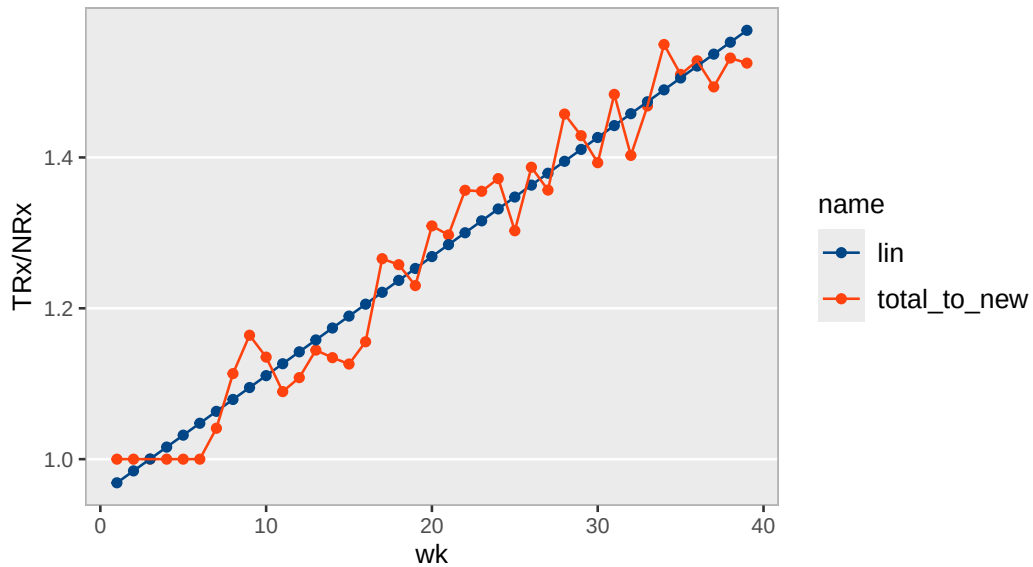
Multiple R-squared: 0.9574, Adjusted R-squared: 0.9562

F-statistic: 830.6 on 1 and 37 DF, p-value: < 2.2e-16

wk	scripts_reported
1	
1	2
5	126
9	163
13	285
17	462
21	515
25	697
29	703
33	878
37	920
2	
2	12
6	154
10	193
14	371
18	483
22	548
26	706
30	840
34	857
38	916
3	
3	35
7	127
11	219
15	384
19	572
23	580
27	677
31	859
35	903
39	944
4	
4	60
8	108
12	205
16	401
20	525
24	616
28	806
32	829
36	897

Tonmya total / new prescriptions linear model

raw Symphony data



Sales Projection - at weekly growth rate - *exponential model*

Formula: $\text{sales} \sim a * (1 + r)^{\text{wk}}$

Parameters:

	Estimate	Std. Error	t value	Pr(> t)	
a	2.770e+02	2.969e+01	9.327	2.96e-11	***
r	4.870e-02	3.613e-03	13.481	7.52e-16	***

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 168.8 on 37 degrees of freedom

Number of iterations to convergence: 12

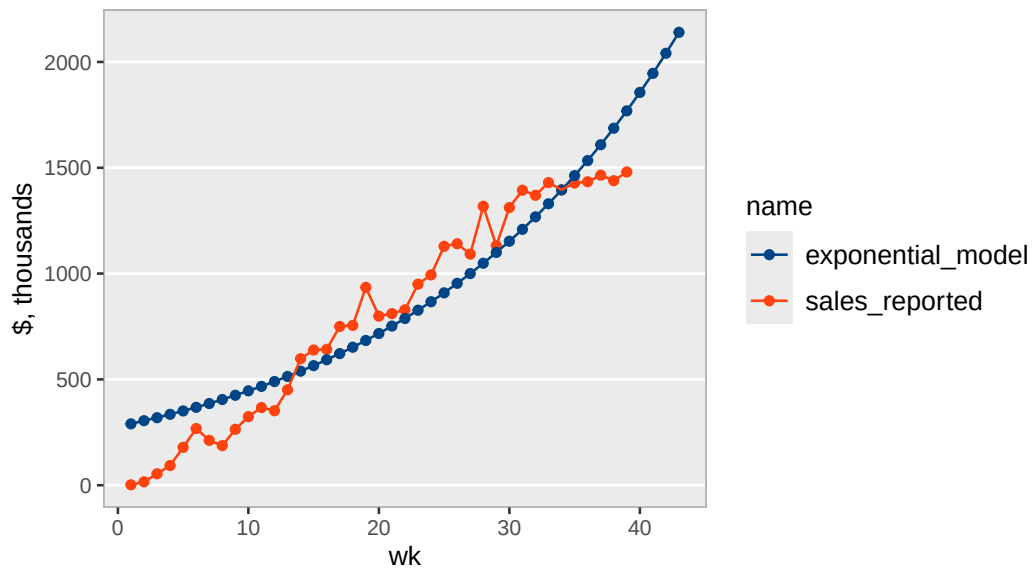
Achieved convergence tolerance: 7.023e-06

Exponential Model

wk	weekly_seq	sales, thousands	exp, thousands	annual, M
1	2025-11-14	\$2	\$290	\$15.1
2	2025-11-21	\$16	\$305	\$15.8
3	2025-11-28	\$54	\$319	\$16.6
4	2025-12-05	\$93	\$335	\$17.4
5	2025-12-12	\$179	\$351	\$18.3
6	2025-12-19	\$268	\$368	\$19.2
7	2025-12-26	\$212	\$386	\$20.1
8	2026-01-02	\$187	\$405	\$21.1
9	2026-01-09	\$264	\$425	\$22.1
10	2026-01-16	\$324	\$446	\$23.2
11	2026-01-23	\$367	\$467	\$24.3
12	2026-01-30	\$352	\$490	\$25.5
13	2026-02-06	\$451	\$514	\$26.7
14	2026-02-13	\$598	\$539	\$28.0
15	2026-02-20	\$639	\$565	\$29.4
16	2026-02-27	\$642	\$593	\$30.8
17	2026-03-06	\$750	\$622	\$32.3
18	2026-03-13	\$755	\$652	\$33.9
19	2026-03-20	\$935	\$684	\$35.5
20	2026-03-27	\$799	\$717	\$37.3
21	2026-04-03	\$811	\$752	\$39.1
22	2026-04-10	\$829	\$788	\$41.0
23	2026-04-17	\$950	\$827	\$43.0
24	2026-04-24	\$994	\$867	\$45.1
25	2026-05-01	\$1,129	\$909	\$47.3
26	2026-05-08	\$1,141	\$954	\$49.6
27	2026-05-15	\$1,092	\$1,000	\$52.0
28	2026-05-22	\$1,318	\$1,049	\$54.5
29	2026-05-29	\$1,132	\$1,100	\$57.2
30	2026-06-05	\$1,312	\$1,153	\$60.0
31	2026-06-12	\$1,394	\$1,209	\$62.9
32	2026-06-19	\$1,370	\$1,268	\$66.0
33	2026-06-26	\$1,430	\$1,330	\$69.2
34	2026-07-03	\$1,396	\$1,395	\$72.5
35	2026-07-10	\$1,428	\$1,463	\$76.1
36	2026-07-17	\$1,434	\$1,534	\$79.8
37	2026-07-24	\$1,465	\$1,609	\$83.7
38	2026-07-31	\$1,439	\$1,687	\$87.7
39	2026-08-07	\$1,480	\$1,769	\$92.0
40	2026-08-14	NA	\$1,856	\$96.5
41	2026-08-21	NA	\$1,946	\$101.2
42	2026-08-28	NA	\$2,041	\$106.1
43	2026-09-04	NA	\$2,140	\$111.3

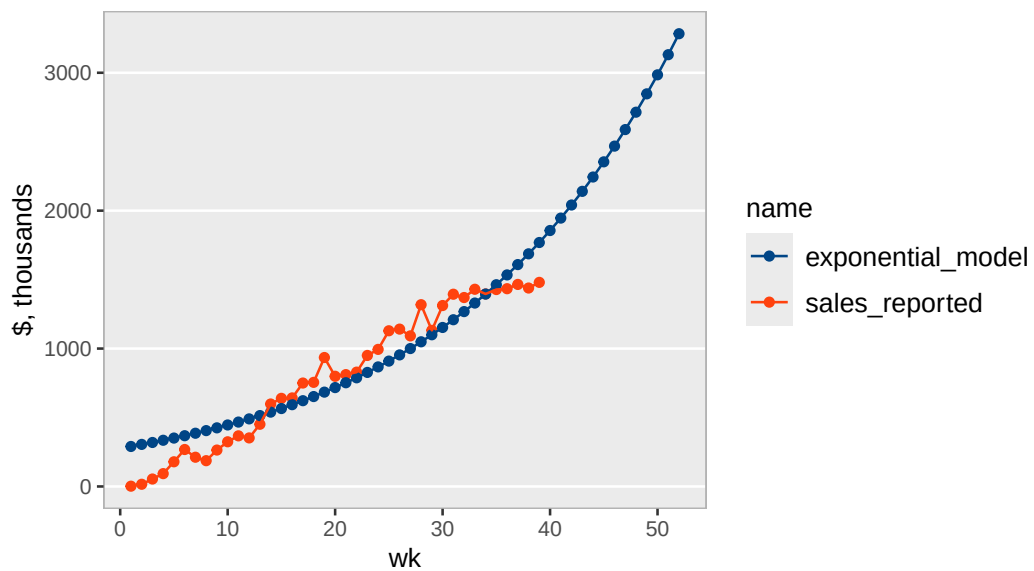
Tonmya sales with exponential model

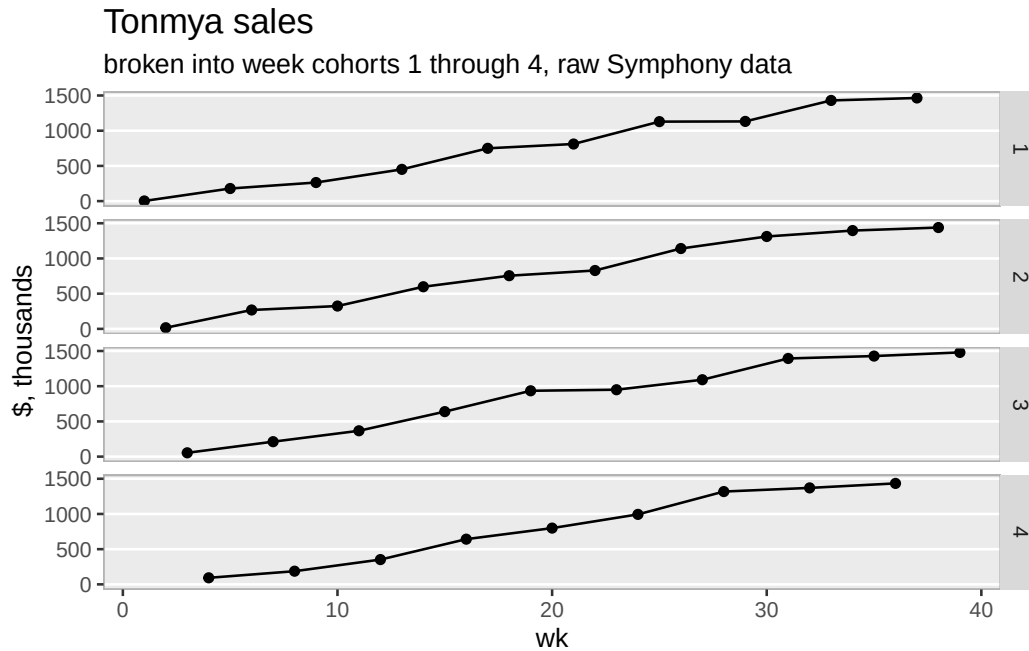
raw Symphony data



Tonmya sales with exponential model

long extrapolation – for amusement, raw Symphony data





Sales Projection - at weekly rate - *linear model*

Call:

```
lm(formula = sales ~ wk, data = df_sales)
```

Residuals:

Min	1Q	Median	3Q	Max
-144.56	-52.23	-13.13	49.55	172.16

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-55.802	24.885	-2.242	0.031 *
wk	43.086	1.084	39.734	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 76.21 on 37 degrees of freedom

Multiple R-squared: 0.9771, Adjusted R-squared: 0.9765

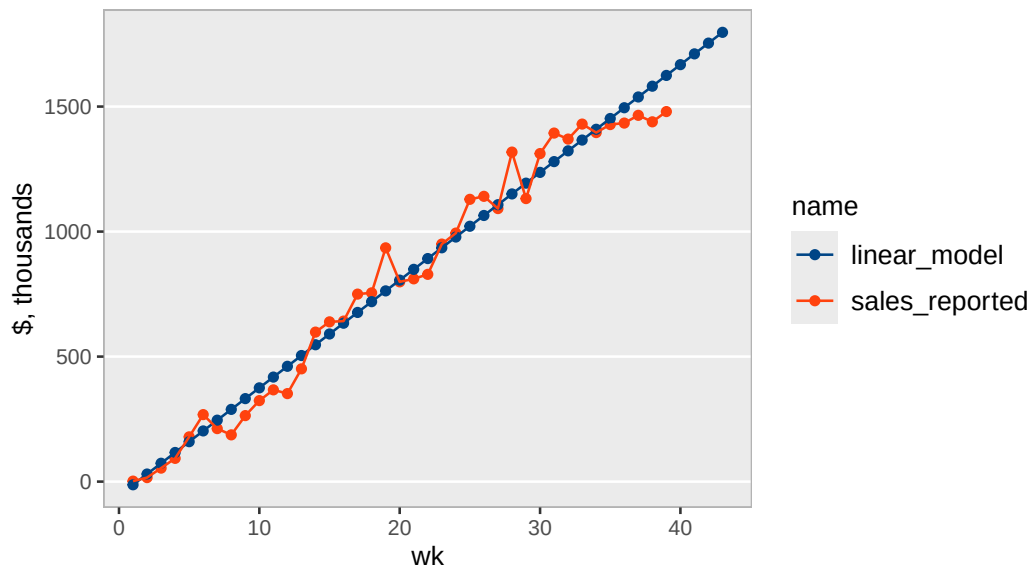
F-statistic: 1579 on 1 and 37 DF, p-value: < 2.2e-16

Linear Model

wk	weekly_seq	sales, thousands	linear, thousands	annual, M
1	2025-11-14	\$2	-\$13	\$2.2
2	2025-11-21	\$16	\$30	\$4.5
3	2025-11-28	\$54	\$73	\$6.7
4	2025-12-05	\$93	\$117	\$9.0
5	2025-12-12	\$179	\$160	\$11.2
6	2025-12-19	\$268	\$203	\$13.4
7	2025-12-26	\$212	\$246	\$15.7
8	2026-01-02	\$187	\$289	\$17.9
9	2026-01-09	\$264	\$332	\$20.2
10	2026-01-16	\$324	\$375	\$22.4
11	2026-01-23	\$367	\$418	\$24.6
12	2026-01-30	\$352	\$461	\$26.9
13	2026-02-06	\$451	\$504	\$29.1
14	2026-02-13	\$598	\$547	\$31.4
15	2026-02-20	\$639	\$590	\$33.6
16	2026-02-27	\$642	\$634	\$35.8
17	2026-03-06	\$750	\$677	\$38.1
18	2026-03-13	\$755	\$720	\$40.3
19	2026-03-20	\$935	\$763	\$42.6
20	2026-03-27	\$799	\$806	\$44.8
21	2026-04-03	\$811	\$849	\$47.1
22	2026-04-10	\$829	\$892	\$49.3
23	2026-04-17	\$950	\$935	\$51.5
24	2026-04-24	\$994	\$978	\$53.8
25	2026-05-01	\$1,129	\$1,021	\$56.0
26	2026-05-08	\$1,141	\$1,064	\$58.3
27	2026-05-15	\$1,092	\$1,108	\$60.5
28	2026-05-22	\$1,318	\$1,151	\$62.7
29	2026-05-29	\$1,132	\$1,194	\$65.0
30	2026-06-05	\$1,312	\$1,237	\$67.2
31	2026-06-12	\$1,394	\$1,280	\$69.5
32	2026-06-19	\$1,370	\$1,323	\$71.7
33	2026-06-26	\$1,430	\$1,366	\$73.9
34	2026-07-03	\$1,396	\$1,409	\$76.2
35	2026-07-10	\$1,428	\$1,452	\$78.4
36	2026-07-17	\$1,434	\$1,495	\$80.7
37	2026-07-24	\$1,465	\$1,538	\$82.9
38	2026-07-31	\$1,439	\$1,581	\$85.1
39	2026-08-07	\$1,480	\$1,625	\$87.4
40	2026-08-14	NA	\$1,668	\$89.6
41	2026-08-21	NA	\$1,711	\$91.9
42	2026-08-28	NA	\$1,754	\$94.1
43	2026-09-04	NA	\$1,797	\$96.3

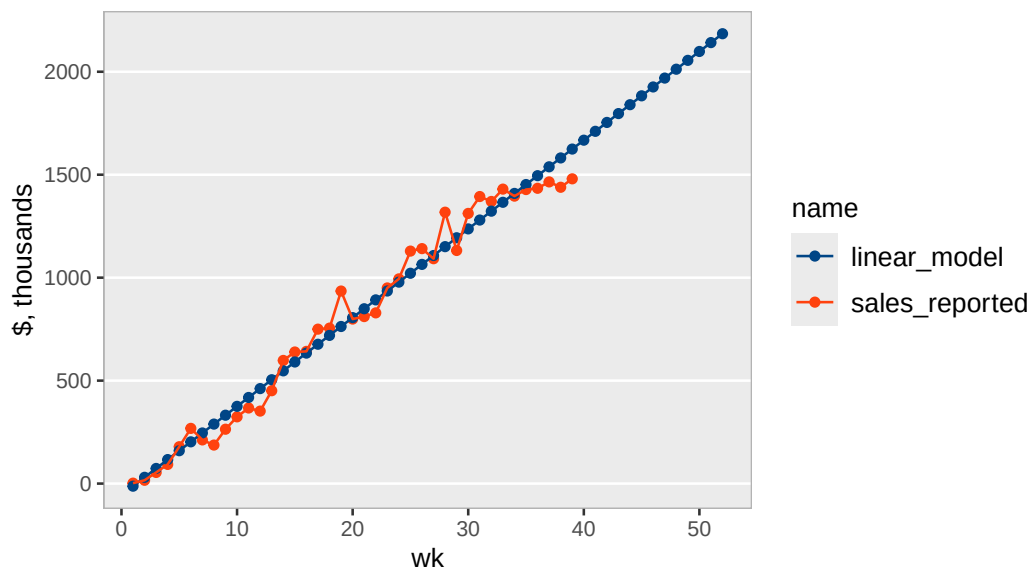
Tonmya sales with linear model

raw Symphony data



Tonmya sales with linear model

long extrapolation, raw Symphony data



Refills - at weekly rate - *linear model*

Call:

```
lm(formula = refills ~ wk, data = trx_nrx_df)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-40.455	-15.137	-3.093	16.909	55.269

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-65.1781	8.0136	-8.133	9.28e-10 ***
wk	9.9089	0.3492	28.377	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.54 on 37 degrees of freedom

Multiple R-squared: 0.9561, Adjusted R-squared: 0.9549

F-statistic: 805.2 on 1 and 37 DF, p-value: < 2.2e-16

